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ECE 2202 – HW 1

Complex Resistive Circuits

1.

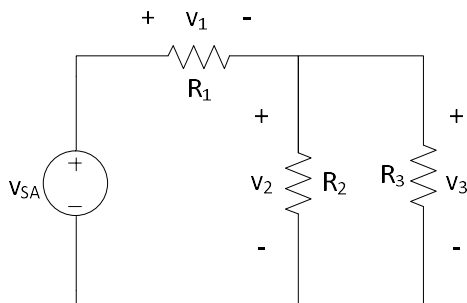


Figure 0.1

For each circuit there is a set of equations.
Mark each equation as true or false.

Figure 0.1

- | | |
|----------|-------------------------|
| <u>F</u> | a) $v_{SA} = v_1$ |
| <u>F</u> | b) $v_{SA} = v_2$ |
| <u>T</u> | c) $v_{SA} = v_1 + v_3$ |

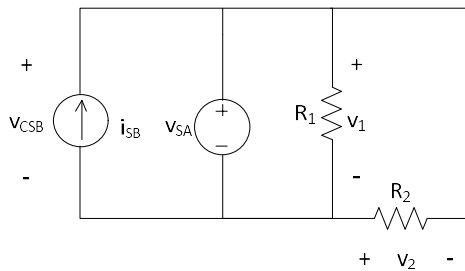


Figure 0.2

Figure 0.2

- | | |
|----------|-----------------------|
| <u>T</u> | d) $v_{SA} = v_1$ |
| <u>T</u> | e) $v_1 = v_2$ |
| <u>F</u> | f) $v_{SA} = v_{CSB}$ |
| <u>T</u> | g) $v_1 = -v_2$ |

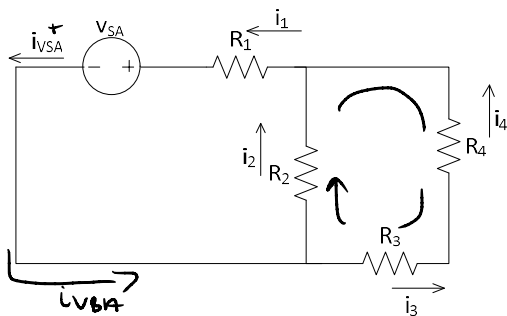
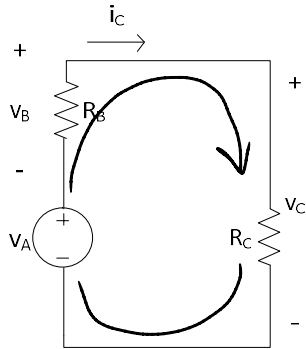


Figure 0.3

Figure 0.3

- | | |
|----------|--------------------------|
| <u>T</u> | h) $i_{VSA} = i_1$ |
| <u>T</u> | i) $i_3 = i_4$ |
| <u>F</u> | j) $i_2 = -i_3$ |
| <u>T</u> | k) $i_2 + i_3 = i_{VSA}$ |

2.



One of the equations below contains a mistake. Write the correct form of the equation and explain why the original equation is incorrect. For this circuit, $v_A = -3.5$ [V].

a) $v_C - v_A - v_B = 0$ ✓

$v_B = R_B i_C$

b) $i_C R_C - (-3.5)[V] - i_C R_B = 0 \rightarrow 3.5[V] =$

KCL

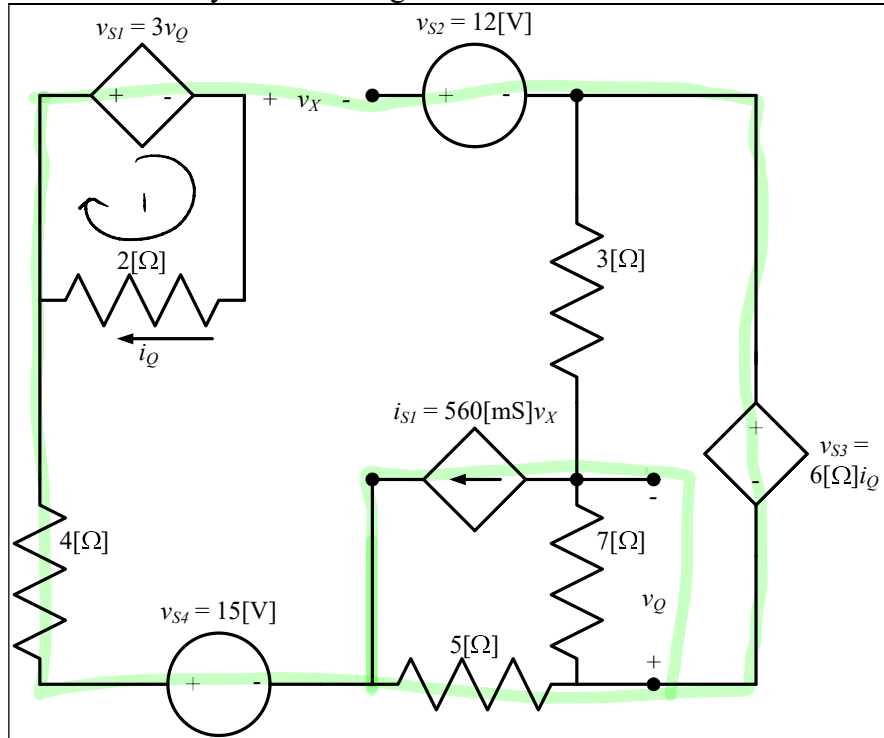
$-v_A + i_C R_B + i_C R_C = 0 \leftarrow \text{correct way}$

3. Using only KVL, KCL, and Ohm's Law (not Node Voltage Method) for the circuit below, solve for the numerical values of the quantities requested.

a) Find $v_X = -6.410$

b) Find the power delivered by the i_{S1} current source. $= 59.1[W]$

c) Find the power absorbed by the v_{S3} voltage source. $= -48.37[W]$



a) Loop 1
 $3v_Q + 2[\Omega]i_Q = 0 \rightarrow i_Q = \frac{3v_Q}{2[\Omega]}$

Loop 2
 $3v_Q + v_X + 12[V] + 6[\Omega]i_Q - (560[mS] \cdot 5[\Omega])v_X - 15[V] = 0$
 $3v_Q - 1.8v_X + \underbrace{6[\Omega]i_Q}_{-9v_Q} - 3[V] = 0 \rightarrow -6v_Q - 1.8v_X = 3[V]$

Loop 3
 $6[\Omega]i_Q + v_Q + 3i_P = 0 \rightarrow -9v_Q + v_Q + \frac{3}{7}v_Q - 1.68v_X = 0$

$$i_P = \frac{v_Q}{7[\Omega]} - 560[mS]v_X$$

$$3i_P = \frac{3}{7}v_Q - 1.68v_X$$

$$\begin{cases} -6v_Q - 1.8v_X = 3[V] \\ -\frac{53}{7}v_Q - 1.68v_X = 0 \end{cases}$$

$$v_Q = 1.4203$$

$$v_X = -6.4010$$

b) power delivered by i_{S1} ← meaning make KVL

$$\begin{aligned} -v + v_Q + 5[\Omega](560[mS]v_X) &= 0 \\ \downarrow \\ 1.420[V] + 2.8(-6.4010) &= 0 \\ v &= -16.5[V] \end{aligned}$$

Power delivered:

$$i_{S1}(V) = P \\ (-16.5[V])(560[mS] \cdot -6.4) = 59.1[W]$$

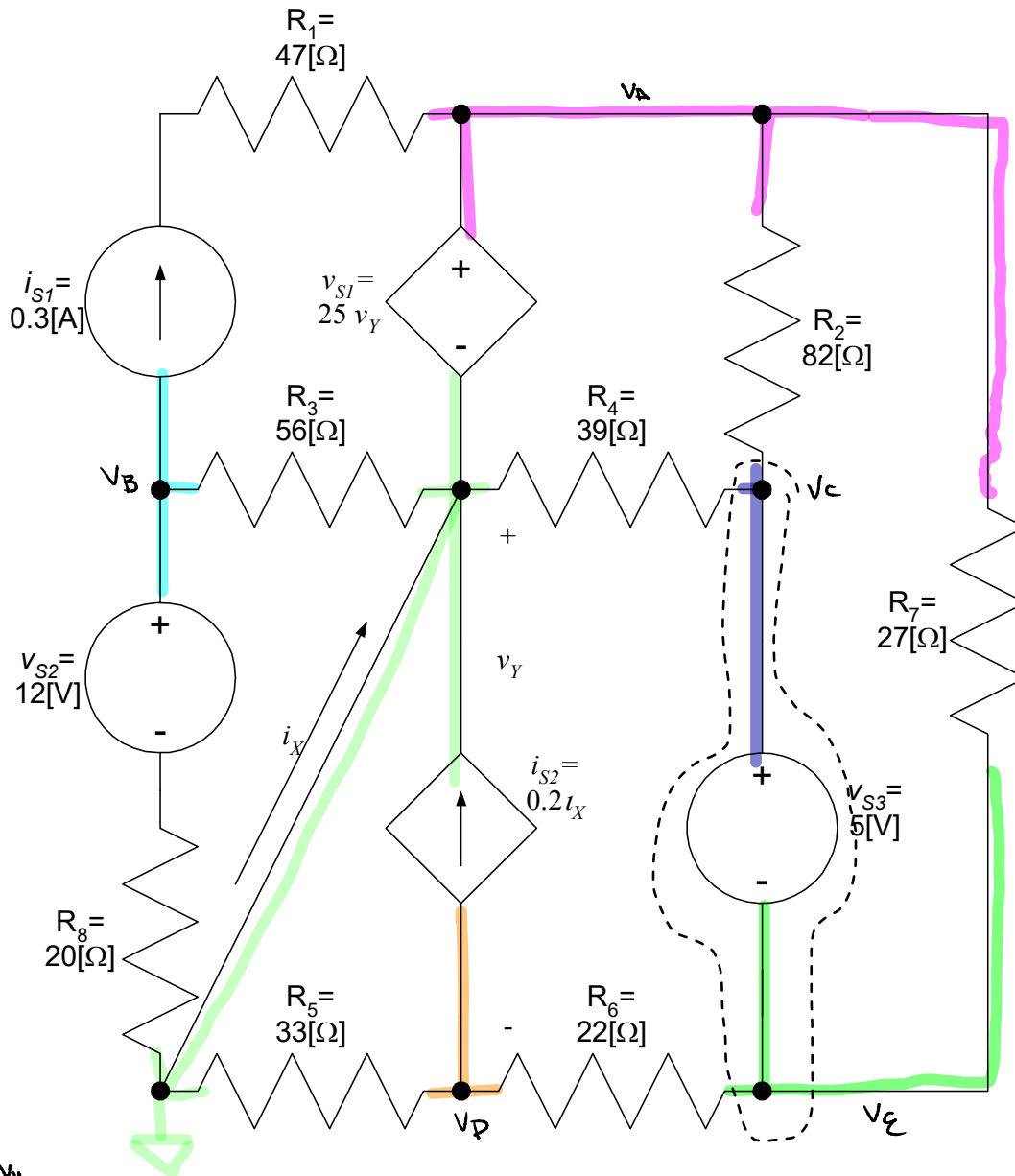
c) Power absorbed: v_{S3}

$$P = V \cdot I = W$$

$$6[\Omega](-2.1305[A])(3.7840) = -48.37[W]$$

$$\begin{aligned} i_Q &= \frac{-3v_Q}{2[\Omega]} = -2.1305[A] \\ i_P &= \end{aligned}$$

4. Use the node voltage method to write a complete set of independent equations that could be used to solve this circuit. Do not attempt to simplify the circuit. Do not solve the circuit – just write out the equations needed to solve. ([Find step-by-step solutions here](#) [PEQWS Module 3 Problem 1](#))



A: $v_A - 25v_Y$

B: $\frac{v_B - 12[V]}{20[\Omega]} + \frac{v_B}{56[\Omega]} - 0.3[A] = 0$

Supernode

(+E:

$v_C + v_E = 5[V]$

$\frac{v_C}{39[\Omega]} + \frac{v_C - v_A}{82[\Omega]} + \frac{v_E - v_D}{22[\Omega]} + \frac{v_E - v_A}{27[\Omega]} = 0$

D: $\frac{v_D - v_E}{22[\Omega]} + \frac{v_D}{33[\Omega]} - 0.2i_X$

$i_X = \frac{12[V] - v_B}{20[\Omega]} - i_X + \frac{v_D}{33[\Omega]} = 0$

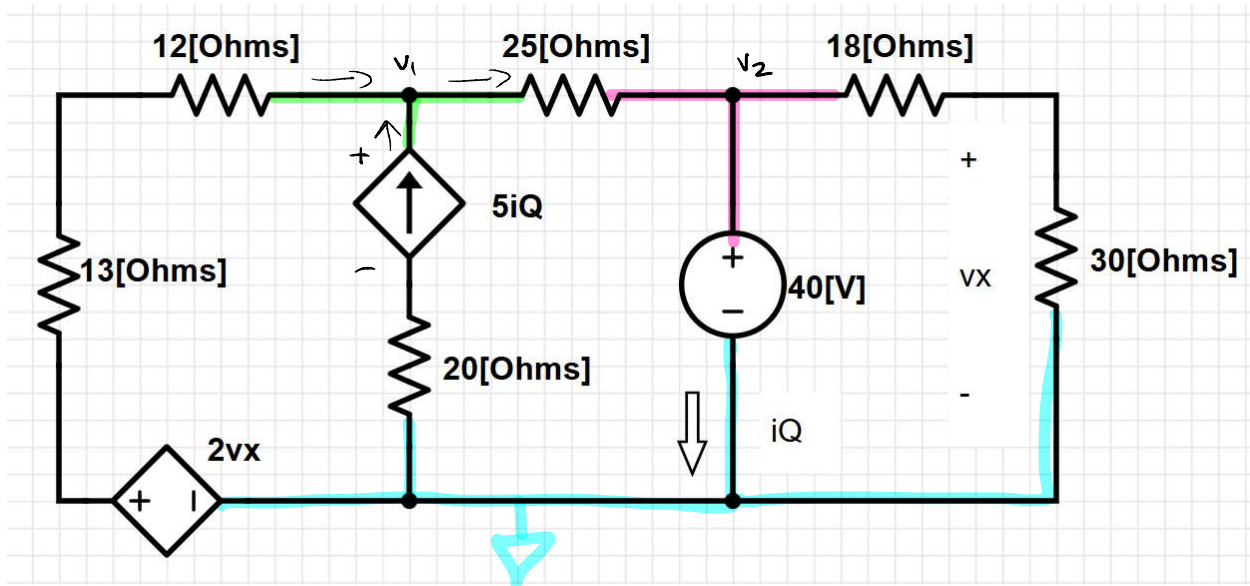
$v_Y = -v_D$

Double check your
set up.

$$V_x = 25[V]$$

$$i_q = 0.422[A]$$

5. Solve the following circuit using node voltage method. Solve for all unknowns labeled on the circuit. Use your CASIO FX-115ES PLUS calculator to solve.



$$V_2 = 40[V] \quad V_x = 40[V] \left(\frac{30}{48} \right) = 25[V]$$

$$\left(\frac{V_1 - 2(25[V])}{25[\Omega]} + \frac{V_1 - 40[V]}{25[\Omega]} - 5i_q = 0 \right)^{25}$$

$$V_1 - 50[V] + V_1 - 40[V] - 125i_q = 0$$

$$2V_1 - 125i_q = 90[V]$$

$$\left(\frac{40[V] - V_1}{25[\Omega]} + i_q + \frac{40[V] - 25[V]}{18[\Omega]} = 0 \right)^{450}$$

$$720[V] - 18V_1 + 450i_q + 375[V] = 0$$

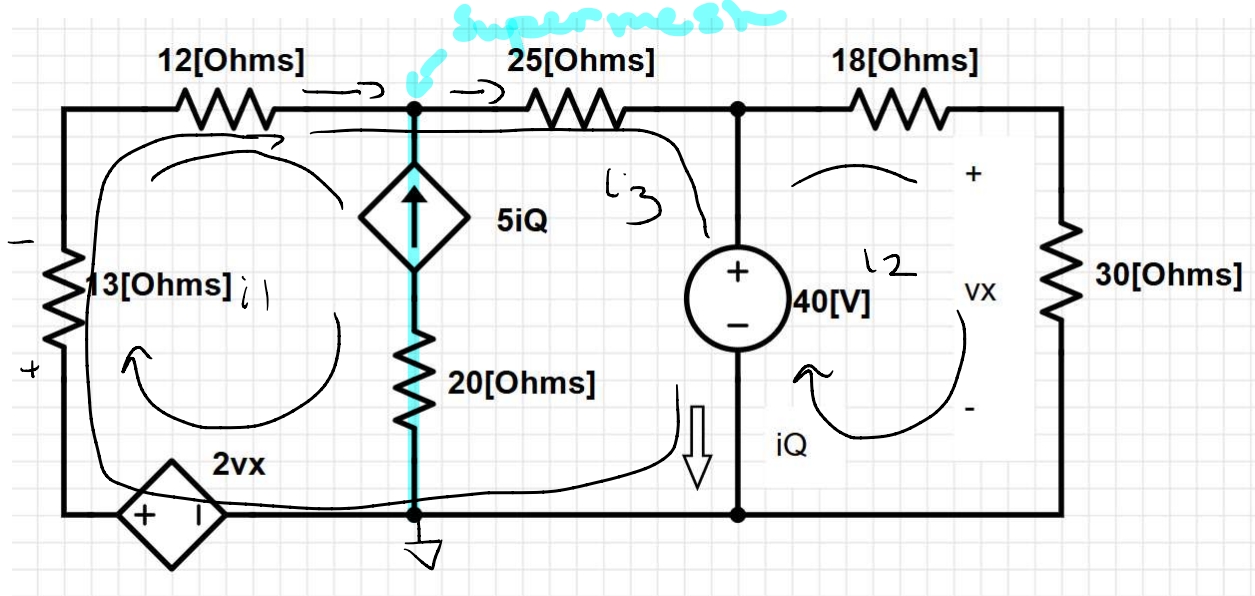
$$-18V_1 + 450i_q + 1095[V] = 0$$

$$V_1 = -71.39[V]$$

$$i_q = 0.422[A]$$

Double check
 i_q

6. Solve the same circuit as the previous question (shown again below) using the mesh current method. Solve for all unknowns in the circuit. Use your CASIO FX-115ES PLUS calculator to solve.



$$\text{loop 1)} -12i_1 - 5i_q - 20(i_1 - i_2) + 2V_x - 13i_1 = 0$$

$$i_q = (i_1 - i_2)$$

$$-12i_1 - 5i_1 - 5i_2 - 20i_1 - 20i_2 + 2V_x - 13i_1 = 0$$

$$-50i_1 - 25i_2 + 2V_x = 0$$

$$\text{loop 2)} -40[V] + 18i_2 + 30i_2 = 0$$

$$48i_2 = 40[V]$$

$$i_2 = 0.833[A]$$

$$V_x = 30i_2 = 30(0.833) = 25[V]$$

$$V_x = 25[V]$$

loop 3

$$-40[V] + 2V_x - 13i_1 - 12i_1 - 25i_2 = 0$$

$$25i_1 - 25i_2 + 2(25) = 40[V]$$

$$50$$

$$25i_1 - 25(0.833) = -10[V]$$

$$+20.825 \quad +20.825$$

$$25i_1 = 10.825 \rightarrow$$

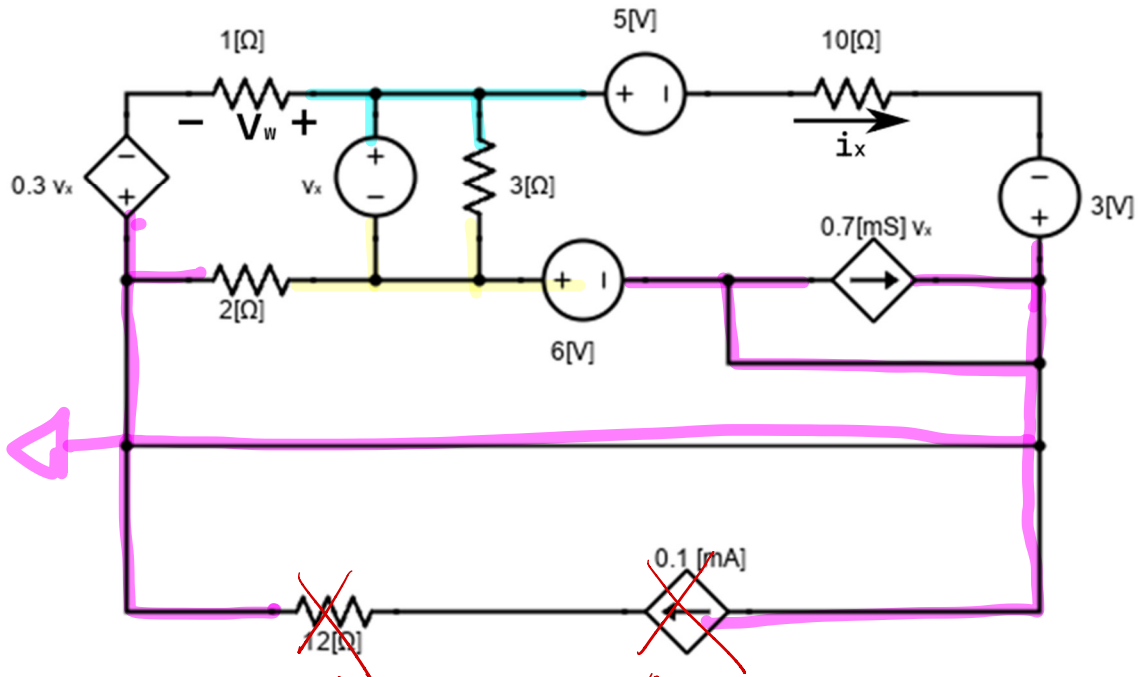
$$i_3 = i_1$$

$$i_q = \frac{i_2 - i_1}{5}$$

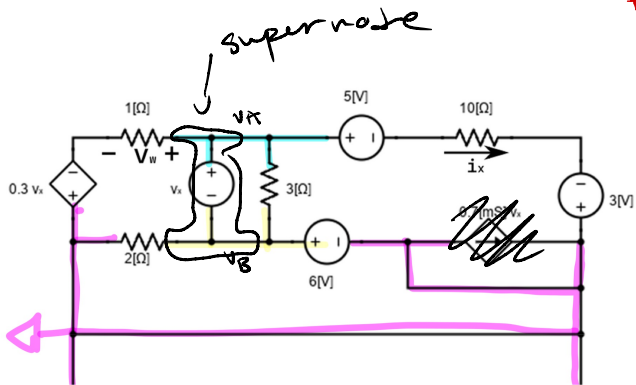
$$\frac{(0.825 - 0.433)[A]}{5} \approx 0.0784[A]$$

$$i_1 = \frac{10.825}{25} = 0.433[A]$$

7. Use the node voltage method to write a complete set of independent equations that could be used to solve this circuit. **You may simplify the circuit.** Solve the circuit and find the values of all unknowns.



↑ these 2 cancel out
cause the are connect
only to ground, mean no
current or voltage is
flowing through this
area.



$$V_B = 6[V]$$

$$V_x = V_A - V_B \rightarrow$$

$$V_A \left(\frac{V_A + 0.3V_A}{1\text{ k}\Omega} \right) + \frac{V_A - 5\text{ V} + 3\text{ V}}{10\text{ k}\Omega} + \frac{6\text{ V}}{2\text{ k}\Omega} = 0$$

$$10V_A + 3V_X + V_A - 2[V] + 30[V] = 0$$

$$11V_A + 3V_X = -28[V]$$

$$-V_A + V_X = -6[V]$$

$$V_w = \frac{V_A + 0.3V_k}{1[\Omega]} \rightarrow 1.303[V]$$

$$i_x = \frac{V_A - 2[V]}{10[\Omega]} = -0.271[A]$$

$$V_A = -0.71 \text{ [V]}$$

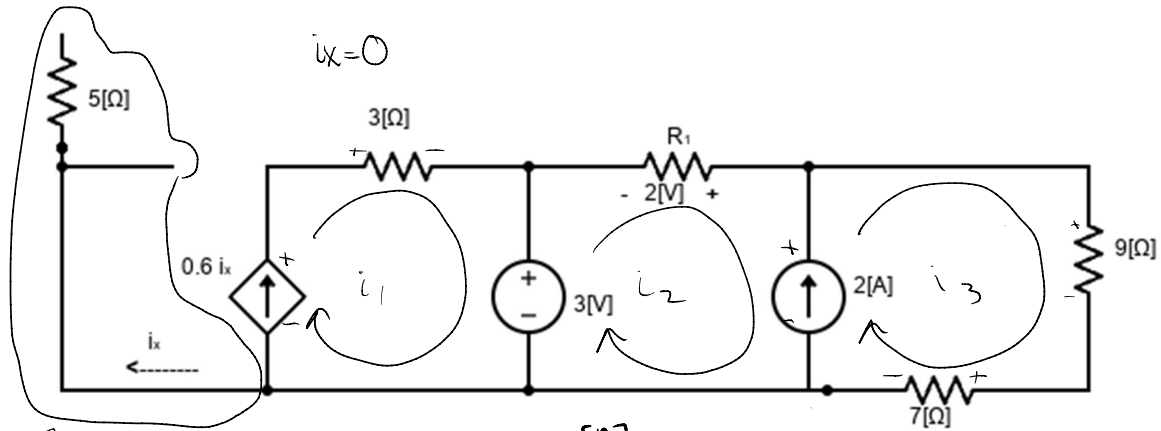
$$V_x = -6.71 [V]$$

$$V_k = -6.71 \text{ [V]}$$

$$V_w = 1.303 [V]$$

$$i_x = -0.271 \text{ [A]}$$

8. Use the mesh current method to write a complete set of independent equations that could be used to solve this circuit. Do not attempt to simplify the circuit. After writing the equations, solve them. Then find i_x and R_1 .



$i_x = 0$
 this is open so no current is flowing through.

$$0.6i_x - 3i_1 - 3[V] = 0$$

Super mesh:

$$-3[V] - 2[V] + 10i_3 = 0$$

$$R_1 i_2 = -2[V] \rightarrow R_1(-1.69) = \frac{-2[V]}{-1.69}$$

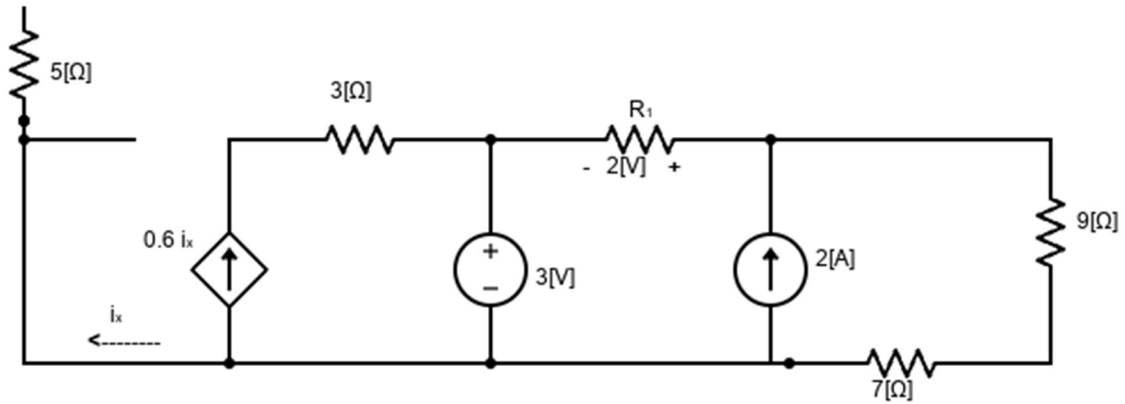
$$i_2 + 2[A] = i_3$$

$$R_1 = 1.18[A]$$

$$i_3 = \frac{5}{10} \rightarrow 0.3125[A]$$

$$i_2 = 0.3125 - 2[A] = -\frac{27}{10}[A]$$

8. Use the mesh current method to write a complete set of independent equations that could be used to solve this circuit. Do not attempt to simplify the circuit. After writing the equations, solve them. Then find i_x and R_1 .



HW 1 Answers

1. a) False b) False c) True d) True e) False f) True g) True h) True i) True j) False k) True
2. a) correct b) $i_C R_C - (-3.5)[V] + i_C R_B = 0$
3. a) $v_X = -6.40[V]$, b) $59.1[W]$, c) $-48.4[W]$
4. See [PEQWS Module 3 Problem 1](#)
5. $v_x = 25 [V]$, $i_Q = 0.422[A]$
6. $v_x = 25 [V]$, $i_Q = 0.422[A]$